

CS4268 Lecture Notes

AY2025/26 Semester 2

CQT and SOC, NUS

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1 Introduction

Quantum computing lies at the intersection of computer science, mathematics, and physics. It utilizes quantum mechanical laws to solve some complex problems (potentially) faster than classical computers. In this course, we shall take the laws of quantum mechanics for granted as our axiomatic "ground truth" and shall focus entirely on their **computational consequences** [NC10].

1.1 Common Computational Tasks

Let's first look at some common computational tasks and their complexities on classical computers.

1.1.1 Multiplication

The classical primary-school (grid) algorithm for multiplying two n -digit numbers performs all digit-wise products explicitly and therefore requires a total of $O(n^2)$ single-digit operations. This was improved by Karatsuba, who introduced a divide-and-conquer strategy that reduces the multiplication of two n -digit numbers to three multiplications of size $n/2$ (with some lower-order additions). Hence we get the recursion

$$T(n) = 3T(n/2) + O(n).$$

yielding an overall time complexity of $O(n^{\log_2 3}) \approx O(n^{1.58})$.

The fastest known classical algorithms for multiplication go even further by utilizing the Fast Fourier Transform (FFT). For example, the Schönhage–Strassen algorithm for integer multiplication achieves a runtime of $O(n \log n \log \log n)$ using FFT-based convolutions.

1.1.2 Factorization

The simplest (and most direct) algorithm for factorizing an integer is *trial division*. It finds a nontrivial divisor by checking integers $d = 2, 3, 4, \dots$ and stopping when $d^2 > N$. This bound works because if $N = ab$ with $a \leq b$, then $a \leq \sqrt{N}$, and if no divisor exists up to $\sqrt{N}$, then N is prime.

In the worst case, this algorithm requires $O(\sqrt{N})$ divisions, or a net $O(\sqrt{N})$ number of arithmetic operations. If the input size is n (where $n = \log_2 N = \text{number of bits}$), then

$$\sqrt{N} = 2^{\frac{1}{2} \log_2 N} = 2^{n/2}.$$

Thus, trial division has a runtime of $O(2^{n/2})$ and is exponential in the bit length of the input. Hence, this algorithm quickly becomes infeasible for practical use as the input size increases.

The most efficient known classical algorithm for factorization is the *General Number Field Sieve* (GNFS). Its runtime is

$$\exp\left(\left(\frac{64}{9}\right)^{1/3} (\log N)^{1/3} (\log \log N)^{2/3}\right),$$

which is sub-exponential in N but still super-polynomial in the input size, making it once again infeasible for practical use when N is large.

RSA

The apparent hardness of factorizing large numbers, especially semiprimes (numbers that are a product of two large primes) is what is utilized by the RSA public key cryptosystems, widely deployed in practice. It exploits the fact that given a huge semiprime N (where $N = pq$ and both p and q are large prime numbers satisfying some sanity checks), it is hard to factorize the number and find p and q .

1.1.3 Unstructured Database Search Problem

In the unstructured database search problem, the idea is that given a bit string $a_1 a_2 \dots a_n$, suppose it is promised that exactly one position contains 1 and rest all are 0s. The goal is to find the unique position i for which $a_i = 1$.

The classical deterministic approach to this problem is the brute force search algorithm. It checks all n possibilities in the input and finds the correct i for which $a_i = 1$ in an overall runtime of $O(n)$. In the deterministic setting, it is easy to see that $\Omega(n)$ is a lower bound as well. Even if we allow classical randomized computation, it can be shown that we still have a complexity of $\Theta(n)$ for this problem.

1.2 Probabilistic Computing

Probabilistic computing consists of randomized algorithms that, in addition to the input, also have access to a source of randomness such as independent and unbiased coin tosses. Due to the source of randomness the algorithm may lead to different execution paths for the same inputs.

Randomized algorithms can usually be divided into two types: Monte Carlo algorithms and Las Vegas algorithms. Monte Carlo algorithms are those that always terminate within a prescribed polynomial time bound but may return an incorrect answer with some bounded probability. Las Vegas algorithms on the other hand always produce a correct output, but their running time is a random variable whose expected value is usually polynomial.

1.2.1 Primality Testing

One of the primary problems where randomized algorithms prove to be useful is primality testing. The goal here is to identify whether the input number is prime or composite. As discussed earlier, deterministic trial division takes $O(2^{n/2})$ time for this.

The Miller-Rabin test, on the other hand, uses Monte Carlo primality testing to achieve a complexity of $O(n^2 \log n \log \log n)$, which is polynomial in the input size. Solovay–Strassen [SS77] is another randomized algorithm that achieves this task in a runtime of $O(k \log^3 n)$, where k is the number of tested bases. It was only very recently (2002) that a deterministic algorithm for primality testing was found. The AKS algorithm [AKS04] provably runs in an overall time of $O(\log^{12} N)$. Subsequent improvements under specific assumptions reduces this to $O(\log^6 N)$ [LP19].

BPP = P

A very strongly held belief in the theoretical computation community is that **BPP = P**. This means that any problem that can be solved in polynomial time with bounded error probability (using randomness) can also be solved in polynomial time deterministically. If true, this would formalize the idea that random bits are merely a convenience for designing faster algorithms, and not a fundamentally stronger computational resource. This remains officially unproven, however.

1.3 Role of Quantum Computing

As discussed earlier, quantum computing utilizes the principles of quantum mechanics to implement algorithms that can solve specific complex problems better than classical computers. Two examples for this are as follows which we shall see in details in this course:

1. **Shor's Algorithm (1994):** Peter Shor discovered a quantum algorithm that can factor integers in **polynomial time**. This gives an exponential speedup than the best known classical counterpart.
 - *Implication:* If we can build a large-scale quantum computer, there is an "easy" way of factorizing the product of 2 large prime numbers effectively breaking the public-key RSA cryptography.
2. **Grover's Algorithm (1996):** Lov Grover discovered a way to solve the unstructured database search problem on a quantum computer in time $O(\sqrt{n})$, which can be proved to be the lower bound as well making its quantum complexity to be $\Theta(\sqrt{n})$, which is a quadratic speedup than its classical counterpart.
 - *Implication:* Its application also extends to a number of other problems, such as the Boolean Satisfiability (SAT) problem. The goal of this problem is to determine whether a Boolean formula with n variables has an assignment that makes it evaluate to true. A deterministic exhaustive search would require testing all 2^n possible assignments, whereas Grover's algorithm can find a satisfying assignment in approximately $2^{n/2}$ steps. Thus, although the runtime remains exponential, the algorithm achieves a quadratic reduction. Consequently, Grover's algorithm does not reduce NP-complete problems to polynomial time, but it effectively halves the exponent in their time complexity.

2 Fundamental Axioms of Quantum Mechanics

In the previous section, we saw that probabilistic computing is "Classical Computing + Randomness." Quantum computing introduces something richer: interference via superposition and entanglement. A high-level summary is:

$$\text{Quantum Computing} = \text{Classical Computing} + \text{Interference (Rotation)}.$$

While probabilistic algorithms introduce randomness via coin tosses, quantum computing uses this extra structure to manipulate information in ways classical computers cannot. This is the source of quantum speedups:

- **Polynomial Speedups:** For problems like unstructured search (Grover's Algorithm).
- **Exponential Speedups:** For problems like factoring (Shor's Algorithm). [\[NC10\]](#)

Since quantum states behave like **vectors** that can be rotated, to understand these algorithms, we must rely heavily on **Linear Algebra**.

2.1 From Bits to Qubits

A classical bit has a state $x \in \{0, 1\}$. A quantum bit, or **qubit**, also has two basis states, which we label using the standard "ket" notation:

$$|0\rangle \quad \text{and} \quad |1\rangle.$$

For now, treat $|0\rangle$ and $|1\rangle$ as abstract labels. However, intuition comes from physics, where objects can be in more than one state at the same time:

- An **electron** may have two energy levels, labeled 0 (ground) and 1 (excited).
- A **photon** may have polarization that is Horizontal or Vertical.

If we label Horizontal as $|0\rangle$ and Vertical as $|1\rangle$, then before measurement, the photon may be in a superposition of both. After measurement, we observe only one outcome.

2.2 Law 1: Superposition

The first fundamental difference is that a qubit can be in a state that is a linear combination of the basis states.

Definition 2.1 (Quantum Mechanics Law 1: Superposition) If a quantum particle can be in one of the 2 basis states $|0\rangle$ and $|1\rangle$, then it can also be in a superposition state:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle,$$

where $\alpha, \beta \in \mathbb{C}$ are complex numbers called **amplitudes**. These amplitudes must satisfy the **normalization condition**:

$$|\alpha|^2 + |\beta|^2 = 1.$$

2.2.1 Geometric Intuition (The Unit Circle)

If we restrict ourselves to **real** amplitudes ($\alpha, \beta \in \mathbb{R}$), the condition $\alpha^2 + \beta^2 = 1$ describes a circle of radius 1. Thus, we can visualize the state of a qubit as a point on the unit circle in $\mathbb{R}^2$.

Example 2.1 Consider the state $|\psi\rangle = 0.8 |0\rangle - 0.6 |1\rangle$. Check normalization:

$$|0.8|^2 + |-0.6|^2 = 0.64 + 0.36 = 1.$$

Geometrically, this corresponds to the point $(0.8, -0.6)$ on the unit circle.

Remark 2.2 (Complex Amplitudes) In the general case, α and β are complex numbers ($a + ib$). The magnitude is defined as $|\alpha| = \sqrt{a^2 + b^2}$. For example, if $|\psi\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{i}{\sqrt{2}} |1\rangle$, then $|\frac{i}{\sqrt{2}}|^2 = \frac{1}{2}$.

2.3 Law 2: Measurement

We cannot "see" the amplitudes α and β directly. We can only extract information via measurement.

Definition 2.2 (Quantum Mechanics Law 2: Measurement) Given a state $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$, if we measure it in the standard basis:

1. We observe outcome $|0\rangle$ with probability $P(0) = |\alpha|^2$.
2. We observe outcome $|1\rangle$ with probability $P(1) = |\beta|^2$.

Crucially, after the measurement, the state **collapses** to the observed outcome.

Example 2.3 Suppose we have the state $|\psi\rangle = 0.8 |0\rangle + 0.6 |1\rangle$.

- **Probability:** We see $|0\rangle$ with probability 0.64, and $|1\rangle$ with probability 0.36.
- **Collapse:** If we see $|0\rangle$, the state *becomes* $|0\rangle$. A second measurement will yield $|0\rangle$ with 100% probability.

This is the key distinction from simply "revealing" a hidden coin toss. In the quantum world, the measurement forces the system to choose a reality.

2.4 The Mathematical Formalism: Qubits as Vectors

So far, we have used $|0\rangle$ and $|1\rangle$ as abstract labels. Mathematically, a quantum state is a vector in a complex vector space $\mathbb{C}^d$. For a single qubit, the dimension is $d = 2$. We identify the basis states with the standard basis vectors:

$$|0\rangle = e_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = e_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Consequently, a general superposition $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ is simply a column vector:

$$|\psi\rangle = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \in \mathbb{C}^2.$$

2.5 Beyond 2 States: Qudits and General Laws

There is nothing special about having exactly two states.

- A **qutrit** ($d = 3$) has basis states $|0\rangle, |1\rangle, |2\rangle$.
- A general **qudit** (d) has basis states $|0\rangle, |1\rangle, \dots, |d-1\rangle$.

In this general setting, the state is a vector in $\mathbb{C}^d$:

$$|\psi\rangle = \alpha_0 |0\rangle + \dots + \alpha_{d-1} |d-1\rangle \iff \begin{bmatrix} \alpha_0 \\ \vdots \\ \alpha_{d-1} \end{bmatrix} \in \mathbb{C}^d.$$

Each basis state $|i\rangle$ is the standard basis vector e_i (a column with a 1 in the i -th position and 0 elsewhere).

We can now restate our two fundamental laws in their full generality.

Definition 2.3 (Law 1 in dimension d : State) A general d -dimensional state is a superposition:

$$|\psi\rangle = \sum_{i=0}^{d-1} \alpha_i |i\rangle, \quad \text{subject to} \quad \sum_{i=0}^{d-1} |\alpha_i|^2 = 1.$$

Definition 2.4 (Law 2 in dimension d : Measurement) Measuring $|\psi\rangle = \sum \alpha_i |i\rangle$ outputs outcome i with probability $|\alpha_i|^2$. Immediately after measurement, the state collapses to $|i\rangle$.

2.5.1 Example: The 4-dimensional system ($d = 4$)

A common example is $d = 4$. We can view this as a single system with 4 levels ($|0\rangle, |1\rangle, |2\rangle, |3\rangle$), or deeply importantly, as a system of **two qubits**. If we have two qubits, the combined system has $2 \times 2 = 4$ basis states:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

Here, $|01\rangle$ means "First qubit is 0, Second qubit is 1." This scaling (2^n dimension for n qubits) is the source of quantum computing's massive state space.

3 Hilbert Space and Quantum States

The language of quantum mechanics is linear algebra. We shall assume some familiarity with it, but introduce a very useful bookkeeping tool – Dirac’s bra-ket notation.

3.1 Hilbert Space

Recall the notion of a Hilbert Space.

Definition 3.1 (Hilbert Space) A **Hilbert Space** $\mathcal{H}$ over $\mathbb{C}$ is a complete vector space over the complex numbers equipped with an inner product $\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ that satisfies:

- Conjugate symmetry: $\langle u, v \rangle = \overline{\langle v, u \rangle}$
- Linearity in the second argument: $\langle u, \alpha v + \beta w \rangle = \alpha \langle u, v \rangle + \beta \langle u, w \rangle$ for $\alpha, \beta \in \mathbb{C}$
- Positive definiteness: $\langle v, v \rangle \geq 0$ with equality if and only if $v = 0$

3.2 Dirac’s Bra-Ket Notation

Dirac’s bra-ket notation:

Definition 3.2 (Bra-ket Notation) A **ket** $|v\rangle$ denotes a column vector in the Hilbert space $\mathcal{H}$.

- Basis vectors are denoted $|0\rangle, |1\rangle, \dots, |n-1\rangle$ for an n -dimensional space.
- A general vector can be written as a linear combination: $|\psi\rangle = \sum_i \alpha_i |i\rangle$ where $\alpha_i \in \mathbb{C}$.
- A **bra** $\langle v|$ denotes the conjugate transpose of $|v\rangle$, i.e., $\langle v| = (|v\rangle)^\dagger$, which is a row vector.
- The inner product of two vectors $|u\rangle$ and $|v\rangle$ is written as $\langle u|v\rangle$.
- Orthonormality of basis vectors: $\langle i|j\rangle = \delta_{ij}$.

Remark 3.1 (Conjugate Transpose as Bra) Given a ket

$$|v\rangle = \begin{pmatrix} v_1 \\ \vdots \\ v_d \end{pmatrix} \in \mathbb{C}^d,$$

its bra is the conjugate transpose

$$\langle v| = (|v\rangle)^\dagger = (v_1^*, \dots, v_d^*).$$

The inner product is therefore

$$\langle u|v\rangle = u^\dagger v,$$

which suppresses explicit matrix multiplication and should be read as “bra- u ket- v ”.

Example 3.2 (Notations for Bases) Dirac’s notation unifies familiar notions of basis vectors:

- **High school basis (in $\mathbb{R}^3$):**

$$\hat{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \hat{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \hat{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

- **Linear algebra basis in $\mathbb{R}^d$ (standard basis):**

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad e_d = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

- **Quantum computational basis in $\mathbb{C}^d$:**

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad |2\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad |d-1\rangle = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

In an n -dimensional quantum system (e.g., qubit: $n = 2$, qutrit: $n = 3$, qudit: $n = d$), these vectors form the *computational basis* $\{|0\rangle, |1\rangle, \dots, |n-1\rangle\}$ and satisfy

$$\langle i | j \rangle = \delta_{ij}.$$

Example 3.3 (Complex Vector Example) Consider the vector

$$|v\rangle = \frac{1}{13} \begin{pmatrix} 3 + 4i \\ 12i \end{pmatrix}.$$

Then the corresponding bra is

$$\langle v| = \frac{1}{13} (3 - 4i, -12i),$$

obtained by taking the conjugate transpose of the ket.

3.3 Quantum States

From now on everything is denoted in bra-ket notation.

In quantum mechanics a system is mathematically modelled by a suitable (to be determined from theory and experiment) Hilbert space. The quantum state representing the system is simply an element of this Hilbert space.

Definition 3.3 (Quantum State) A **quantum state** $|\psi\rangle$ is an element of a Hilbert space $\mathcal{H}$ with unit norm, i.e., $\langle \psi | \psi \rangle = 1$.

For a quantum system with orthonormal basis $\{|0\rangle, |1\rangle, \dots, |n-1\rangle\}$, any quantum state can be written as a superposition:

$$|\psi\rangle = \sum_{i=0}^{n-1} \alpha_i |i\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle + \dots + \alpha_{n-1} |n-1\rangle$$

where $\alpha_i \in \mathbb{C}$ are complex amplitudes satisfying the normalization condition $\sum_{i=0}^{n-1} |\alpha_i|^2 = 1$.

In column vector representation:

$$|\psi\rangle = \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{n-1} \end{pmatrix}.$$

Physical Interpretation: When we measure the quantum state $|\psi\rangle$ in the computational basis, the probability of observing outcome i is $|\alpha_i|^2$. This is **Born's rule**.

Example 3.4 (Quantum States as Superpositions) For a single qubit (2-dimensional quantum system):

- Computational basis states: $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- Superposition state: $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
- Another superposition: $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$
- General qubit state: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ where $|\alpha|^2 + |\beta|^2 = 1$

The states $|+\rangle$ and $|-\rangle$ form the Hadamard basis (also called the X -basis or plus-minus basis).

Of course, we know the basis of a (nontrivial) Hilbert space is not unique. We could choose to represent states w.r.t. different bases, so the corresponding column vectors would have different physical interpretations.

4 Measuring w.r.t. different bases and the Elitzur-Vaidman Test

4.1 Measurement in Different Bases

Up until now, we have only considered measurements with respect to the standard basis ($\{|0\rangle, |1\rangle\}$ in the 2 dimensional case). However, we are not restricted to just measuring in the standard basis and can use any valid set of orthonormal basis states as the measurement basis. Let us now consider the problem of making measurements with respect to different bases.

Example 4.1 (Measurement with respect to Different Bases) Consider a qubit in state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

Measurement w.r.t. computational basis $\{|0\rangle, |1\rangle\}$:

- $\Pr(\text{measure } |0\rangle) = |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2},$
- $\Pr(\text{measure } |1\rangle) = |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}.$

Measurement w.r.t. Hadamard basis $\{|+\rangle, |-\rangle\}$:

First, express $|+\rangle$ in the new basis. Since $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$, we have:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \left[\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) + \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle) \right] = 1 \cdot |+\rangle + 0 \cdot |-\rangle.$$

Thus

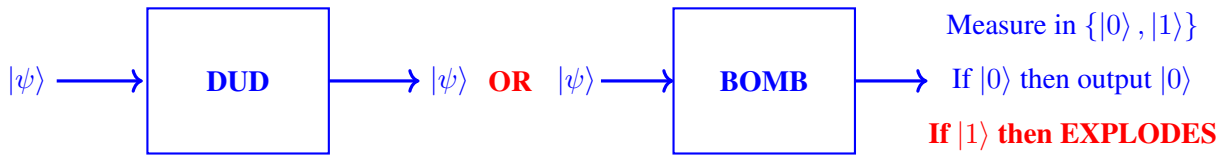
- $\Pr(\text{measure } |+\rangle) = |1|^2 = 1,$

- $\Pr(\text{measure } |-\rangle) = |0|^2 = 0$.

Key insight: When we measure $|+\rangle$ in the Hadamard basis $\{|+\rangle, |-\rangle\}$, the coefficient of $|+\rangle$ is 1 and the coefficient of $|-\rangle$ is 0. This makes intuitive sense: if the system is already in state $|+\rangle$ and we measure in the $\{|+\rangle, |-\rangle\}$ basis, we are guaranteed to get the outcome $|+\rangle$. Thus, the same quantum state gives different measurement outcomes depending on the measurement basis.

4.2 Elitzur-Vaidman Bomb-Tester

Now let us consider the following thought experiment. Suppose we are given a box that is either a DUD or a bomb. We want to figure out which box we are given without triggering an explosion. The behaviour of the two possible boxes is shown below:



Stay alive and figure out which box we are given.

Some preliminary options that comes to mind are sending $|0\rangle$ or $|+\rangle$ through the unknown box, and then measure the output from the box. Let us examine what happens in each of these cases.

Classical Strategy:

If we send $|0\rangle$, we are unable to identify if it is a dud or a bomb as the output is the same. If we send $|1\rangle$:

- No explosion $\Rightarrow$ it's a dud!
- Bomb explodes $\Rightarrow$ it's a bomb! (but you dead)

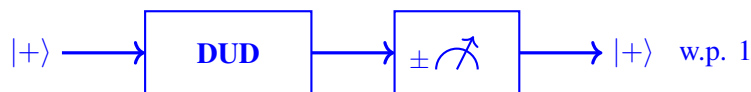
Conclusion: Classically impossible to distinguish dud/bomb without triggering an explosion.

Preliminary Quantum Strategy:

Send in $|\psi\rangle = |+\rangle$ and measure w.r.t. $\{|+\rangle, |-\rangle\}$.

Case 1: Dud bomb

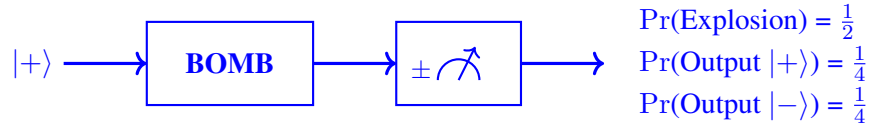
If it's a dud, we will always see $|+\rangle$ because 100% of the amplitude is on $|+\rangle$.



Case 2: Working bomb

Three possible outcomes:

- Bomb measures $|0\rangle$ and explodes (with probability $\frac{1}{2}$).
- Bomb measures $|1\rangle$ and does not explode:
 - We observe $|+\rangle$ (with probability $\frac{1}{4}$): inconclusive (could be dud or working)
 - We observe $|-\rangle$ (with probability $\frac{1}{4}$): Working bomb identified!



If we observe $|-\rangle$, we know it is a bomb because: with a dud, we get $|+\rangle$ all the time (with probability 1).

Conclusion: $\frac{1}{4}$ success rate for identifying working bomb without explosion.

Can we improve upon this? Yes. First, but first let us make a brief detour into the notion of rotations (more generally, unitaries later on) on quantum states.

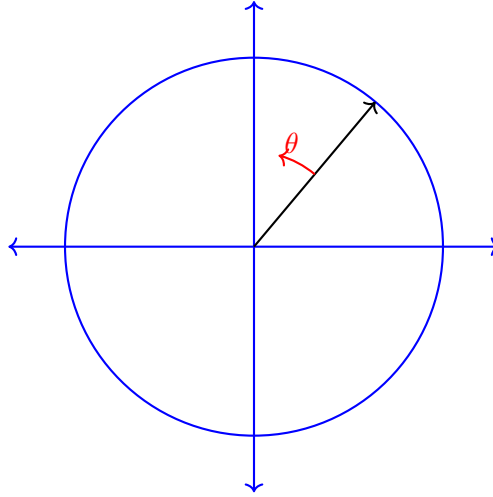
4.3 Rotations and Reflections

4.3.1 Rotations

Definition 4.1 (Rotations on a Single Qubit) For any θ , we can build a physical device that rotates a qubit by angle θ . Consider a transformation that rotates the quantum state by a small angle θ . Starting from $|0\rangle$, we apply a rotation: $R(\theta) : |0\rangle \mapsto \cos(\theta) |0\rangle + \sin(\theta) |1\rangle$

$$|0\rangle \rightarrow \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

$$|1\rangle \rightarrow \begin{pmatrix} \cos(\frac{\pi}{2} + \theta) \\ \sin(\frac{\pi}{2} + \theta) \end{pmatrix} = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$$



For a general state $\alpha |0\rangle + \beta |1\rangle$, it gets transformed to:

$$\begin{aligned} \alpha |0\rangle + \beta |1\rangle &\rightarrow \alpha \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} + \beta \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \end{aligned}$$

where $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ goes to the vector $\begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ goes to the vector $\begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$.

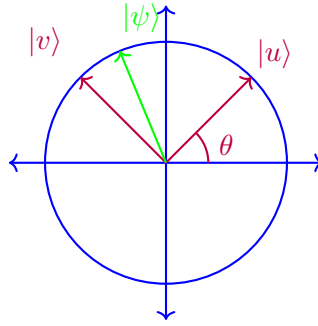
Any state $|\psi\rangle$ gets transformed to $R_\theta |\psi\rangle$, where:

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Note:

- α and β may be complex numbers.
- Viewing R_θ as a rotation may only be intuitive for real amplitudes.

Measuring w.r.t. $|u\rangle, |v\rangle$ basis via rotation:



$$|u\rangle = R_\theta |0\rangle$$

$$|v\rangle = R_\theta |1\rangle$$

For any state:

$$\begin{aligned} |\psi\rangle &= \alpha |u\rangle + \beta |v\rangle \\ &= \alpha \cdot R_\theta |0\rangle + \beta \cdot R_\theta |1\rangle \end{aligned}$$

How to measure in $|u\rangle, |v\rangle$ while actually measuring in standard basis?

Apply $R_{-\theta}$ to $|\psi\rangle$:

$$R_{-\theta} |\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

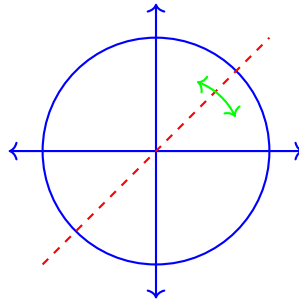
Then measure in $\{|0\rangle, |1\rangle\}$:

- $|0\rangle$ with probability $|\alpha|^2 \rightarrow$ output $|u\rangle$
- $|1\rangle$ with probability $|\beta|^2 \rightarrow$ output $|v\rangle$.

4.3.2 Reflections

We can also build reflectors along any axis.

Example 1: Reflection along diagonal axis



Reflection transformations:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|0\rangle \rightarrow |1\rangle, \quad |1\rangle \rightarrow |0\rangle$$

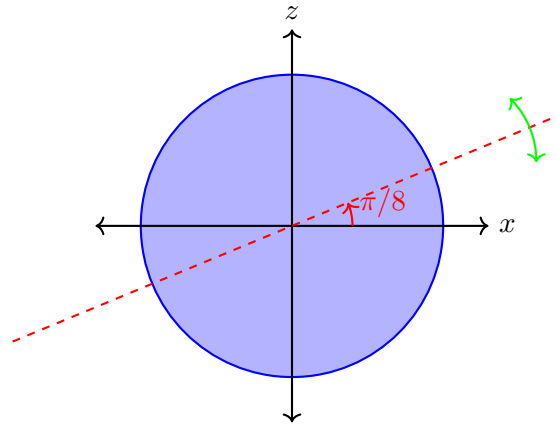
For a general state:

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} \text{ gets transformed to } \begin{pmatrix} \beta \\ \alpha \end{pmatrix}$$

NOT operation:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \text{ maps to } \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Example 2: Hadamard gate (reflection along different axis)



Hadamard transformations:

$$|0\rangle \text{ maps to } |+\rangle$$

$$|1\rangle \text{ maps to } |-\rangle$$

$$\psi = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \text{ is mapped to } \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

This is labeled the Hadamard gate and is arguably the most important transformation in quantum computing.

4.4 Elitzur-Vaidman Revisited

Improved Quantum Strategy via small-angle rotations (Quantum Zeno effect) We now show how to make the explosion probability arbitrarily small. Let $\varepsilon = \frac{\pi}{2N} > 0$ be a small angle, i.e. N is large. Define the rotation operator

$$R_\varepsilon = \begin{pmatrix} \cos \varepsilon & -\sin \varepsilon \\ \sin \varepsilon & \cos \varepsilon \end{pmatrix}.$$

Starting from $|0\rangle$, we perform the following protocol:

1. Apply R_ε .
2. Send the qubit through the unknown box.
3. If no explosion occurs, repeat the above steps N times.
4. Finally, measure in the computational basis.

Dud: Since $N = \frac{\pi}{2\varepsilon}$, in the dud case the cumulative rotation is

$$R_\varepsilon^N = R_{\pi/2},$$

which maps $|0\rangle$ close to $|1\rangle$. Thus after N rounds the state is approximately $|1\rangle$, and a final measurement yields outcome $|1\rangle$ with full certainty.

Working Bomb: After a single application of R_ε the state becomes $\cos \varepsilon |0\rangle + \sin \varepsilon |1\rangle$. The box then measures in the computational basis. The probability of explosion in that round is $\sin^2 \varepsilon \leq \varepsilon^2$. If no explosion occurs, the state collapses back to $|0\rangle$, and the process repeats.

By the union bound, the probability of at least one explosion over N rounds is upper-bounded by $N\varepsilon^2 = \frac{\pi}{2}\varepsilon$, tending to 0 as $\varepsilon \rightarrow 0$. Meanwhile, if no explosion occurs for any of the N rounds, the final measurement yields $|0\rangle$, in stark contrast to the dud case, which yields $|1\rangle$. Thus by choosing ε sufficiently small, we can distinguish a live bomb from a dud with arbitrarily high confidence, while making the probability of explosion arbitrarily small.

In a nutshell, using this quantum Zeno strategy we have

$$\begin{aligned} \Pr(\text{observe } |1\rangle \mid \text{DUD}) &= 1 \\ \Pr(\text{observe } |0\rangle \mid \text{BOMB}) &\geq 1 - \frac{\pi}{2}\varepsilon. \end{aligned}$$

5 Unitary Operations

We have already seen some operations like the **NOT** operation and the **Hadamard** operation. We will now understand what kind of transformations are allowed on quantum states. The transformations that are physically allowed on quantum states are called unitary transformations (or operations). Intuitively, unitaries preserve the ‘length’ of quantum states, thus preserving probability densities.

Quantum Mechanics Law 3: Quantum states can be transformed (without losing information) via **unitary transformations**.

Definition 5.1 (Unitary Matrices) A linear transformation $U : \mathcal{H} \rightarrow \mathcal{H}$ on a complex Hilbert space is called **unitary** if it preserves the norm of every state. That is,

$$\|U|\psi\rangle\|^2 = \|\psi\rangle\|^2 \quad \text{for all } |\psi\rangle \in \mathcal{H}.$$

Here are a few examples of unitary matrices:

Example 5.1 (Examples of Unitary Matrices)

1. **Identity:** $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

2. **Pauli-X (NOT gate):** $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

Effect: $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$ (bit flip)

3. **Pauli-Y:** $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

4. **Pauli-Z (phase flip):** $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Effect: $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$

5. **Hadamard gate:** $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Effect: Creates equal superpositions

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

Note: $H^2 = I$ (Hadamard is self-inverse).

6. **Phase gate (S gate):** $S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$

7. **T gate ($\pi/8$ gate):** $T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$

8. **Rotation gates:** $R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$

5.1 Properties of Unitaries

Let us present the various characterizations of unitary matrices.

Proposition 5.1 (Basic Properties of Unitary Matrices)

1. **Preservation of inner products and norms:** for any $|v\rangle, |w\rangle$: $\langle Uv|Uw\rangle = \langle v|U^\dagger U|w\rangle = \langle v|w\rangle$. In particular, $\|U|v\rangle\|^2 = \| |v\rangle \|^2$.
2. **Preservation of angles:** Since $\cos(\theta) := \frac{\langle v|w\rangle}{\|v\|\|w\|}$ and unitary transformations preserve both inner products and norms, the angle between any two vectors is also preserved.
3. **Eigenvalues lie on unit circle:** If $U|v\rangle = \lambda|v\rangle$, then $|\lambda| = 1$. *Proof:* $\| |v\rangle \|^2 = \|U|v\rangle\|^2 = \|\lambda|v\rangle\|^2 = |\lambda|^2\| |v\rangle \|^2$, so $|\lambda| = 1$.
4. **Determinant has unit modulus:** $|\det(U)| = 1$. *Proof:* take determinants on both sides of $U^\dagger U = I$.
5. **Composition of unitaries is unitary:** If U, V are unitary, then $(UV)^\dagger(UV) = V^\dagger U^\dagger UV = V^\dagger V = I$.
6. **Inverse is unitary:** If U is unitary, so is $U^\dagger = U^{-1}$.

7. **Reversibility:** Quantum evolution is reversible. Given $|\psi'\rangle = U|\psi\rangle$, we can always recover $|\psi\rangle = U^{-1}|\psi'\rangle = U^\dagger|\psi'\rangle$.
8. **Square roots exist:** Every unitary U has a square root W , which is also unitary, such that $W^2 = U$. This allows for fractional quantum operations.

Example: Consider the NOT gate (Pauli-X): $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Its square root is:

$$\sqrt{X} = \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}$$

We can verify: $\sqrt{X} \cdot \sqrt{X} = \frac{1}{4} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X$

On the Bloch sphere, $\sqrt{X}$ represents a rotation halfway to the full NOT operation.

5.2 Classical Analogue of Unitary Transformations

It is instructive to compare unitary evolution in quantum mechanics with the way probability distributions evolve in classical randomized processes. In the classical setting, a system with d possible outcomes can be represented by a probability vector

$$p = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_d \end{bmatrix}, \quad p_i \geq 0, \quad \sum_{i=1}^d p_i = 1.$$

A randomized procedure transforms p into a new distribution p' via multiplication by a matrix M :

$$p' = Mp.$$

To preserve total probability, such a matrix must be column-stochastic, meaning its entries are non-negative and each column sums to 1:

$$M_{ij} \geq 0 \quad \text{and} \quad \sum_{i=1}^d M_{ij} = 1 \quad \text{for all } j.$$

We now illustrate this with a classical example.

Consider the following random process: we roll a fair die; if the outcome is odd, we toss a fair coin; if it's heads, add 1 to the outcome of the fair die.

Let the state space be $\{1, 2, 3, 4, 5, 6\}$, and start from the uniform distribution

$$p = \begin{bmatrix} \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \\ \frac{1}{6} \end{bmatrix}.$$

The transformation implemented by the above procedure is captured by the column-stochastic matrix

$$M = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 1 \end{bmatrix}, \quad p' = Mp.$$

Starting from an odd face 1, 3, 5, the coin toss either keeps the value (tails) or moves it to the next even face (heads), giving the $\frac{1}{2}$ - $\frac{1}{2}$ split in the corresponding column. Starting from an even face 2, 4, 6, the value stays unchanged, which is why the corresponding column has a single 1.

This classical picture is quite similar to quantum evolution in that both are linear transformations on vectors. However, there are key differences.

First, classical stochastic evolution preserves the ℓ_1 norm,

$$\|p'\|_1 = \|p\|_1 = 1,$$

whereas quantum evolution preserves the ℓ_2 norm,

$$\|U|\psi\rangle\|_2 = \|\psi\|_2.$$

Second, classical stochastic transformations are generally not reversible: a column-stochastic matrix need not be invertible, and even when it is invertible, its inverse typically does not have non-negative entries and therefore does not correspond to a valid stochastic process. In contrast, every unitary matrix is reversible, with inverse $U^\dagger$ as seen in the characterisation. This reversibility is one of the fundamental distinctions between classical and quantum dynamics.

6 Composite Systems

We have discussed how to describe a single quantum system via the notion of quantum states residing in a Hilbert space, how they can be transformed and how information is extracted from the system via measurements. Now we discuss how to characterize *composite* systems. Given two (or more) quantum systems, each with its own Hilbert space, how do we describe them collectively as a whole? In particular, if we only effect changes on a component of this composite system, how can we describe this mathematically as an operation on the entire system?

This is achieved via the notion of *tensor products*.

6.1 Tensor Products

Let us consider a system comprising simply two qudits. This system is mathematically described by the tensor product of the Hilbert spaces of the individual qudits:

Definition 6.1 (Tensor Product of Hilbert Spaces) Given two qudit Hilbert spaces $\mathcal{H}_1 = \mathbb{C}^{d_1}$, $\mathcal{H}_2 = \mathbb{C}^{d_2}$ spanned respectively by the computational basis vectors

$$\mathcal{B}_1 = \{|0\rangle, |1\rangle, \dots, |d_1 - 1\rangle\}, \quad \mathcal{B}_2 = \{|0\rangle, |1\rangle, \dots, |d_2 - 1\rangle\},$$

the **tensor product** of $\mathcal{H}_1$ and $\mathcal{H}_2$ is defined as the Hilbert space

$$\mathcal{H}_1 \otimes \mathcal{H}_2 = \text{span}\{|i\rangle \otimes |j\rangle : 0 \leq i \leq d_1 - 1, 0 \leq j \leq d_2 - 1\}.$$

Each (normalized) state $|\psi\rangle = \sum_{ij} a_{ij} |i\rangle \otimes |j\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$, $\sum_{ij} |a_{ij}|^2 = 1$, is a legitimate characterization of the entire system. Also, for brevity we shall henceforth write $|\alpha\rangle \otimes |\beta\rangle$ simply as $|\alpha\beta\rangle$.

More generally, for $n > 2$ systems, the tensor product generalizes naturally to

$$\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \dots \otimes \mathcal{H}_n = \text{span}\{|i_1\rangle \otimes \dots \otimes |i_n\rangle\}.$$

Remark 6.1 (Existence and Uniqueness of Tensor Products) Note that above we have defined the tensor product as being the span of the *formal* symbols $|i\rangle \otimes |j\rangle$ – we have not at all dissected what the symbols $|i\rangle \otimes |j\rangle$ really mean. In particular, what is the difference between $|i\rangle \otimes |j\rangle$ and the *direct sum* $(|i\rangle, |j\rangle)$? Superficially, both seem quite similar in the sense that the individual constituents are simply concatenated together (albeit in different notations). Indeed, these two notions are closely related: roughly speaking, we think of the symbol $\otimes$ as a *bilinear map* from the direct sum/Cartesian product $H_1 \times H_2$ to the Hilbert space Z (we use the symbol Z because we have not yet proved that this thing exists). That is, the purported map $\otimes$ behaves as such (on all vectors $|u\rangle, |v\rangle$, not just the basis vectors):

$$\otimes(|u\rangle, |v\rangle) := |u\rangle \otimes |v\rangle$$

such that

$$(a|u_1\rangle + b|u_2\rangle) \otimes |v\rangle = a|u_1\rangle \otimes |v\rangle + b|u_2\rangle \otimes |v\rangle \quad (6.1)$$

$$|u\rangle \otimes (a|v_1\rangle + b|v_2\rangle) = a|u\rangle \otimes |v_1\rangle + b|u\rangle \otimes |v_2\rangle. \quad (6.2)$$

The existence and (essential) uniqueness of the map $\otimes$ is nontrivial to show. Once these are taken care of we can start calling Z as $H_1 \otimes H_2$. If you are interested, the wiki article on [tensor products](#) is a good starting point. Be warned however that this leads to a long rabbit hole.

That was a lot of yapping – but we won't worry about any of that in this course. In practice, think of $\otimes$ as concatenation plus multilinearity: the symbol $\otimes$ is used to stack different basis vectors from different Hilbert spaces together, and that $\otimes$ is linear with respect to *all* its arguments, i.e. [Eq. \(6.1\)](#)

Fact 6.1 (Column Vector Representation of Tensor Products) Given vectors $|\alpha\rangle \in \mathcal{H}_1$ and $|\beta\rangle \in \mathcal{H}_2$, the column representation of $|\alpha\rangle \otimes |\beta\rangle$ is given by

$$|a\rangle \otimes |b\rangle = \begin{pmatrix} a_0 \\ \vdots \\ a_{d_1-1} \end{pmatrix} \otimes \begin{pmatrix} b_0 \\ \vdots \\ b_{d_2-1} \end{pmatrix} = \begin{pmatrix} a_0 b_0 \\ \vdots \\ a_0 b_{d_2-1} \\ \vdots \\ a_{d_1-1} b_0 \\ \vdots \\ a_{d_1-1} b_{d_2-1} \end{pmatrix}.$$

In particular, the tensor product of the basis vectors is given by

$$|i\rangle \otimes |j\rangle = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

where the 1 appears at position $i \cdot d_2 + j$.

These results are immediate from the definition of the tensor product.

Example 6.2 For two qubits ($d_1 = d_2 = 2$), the computational basis states are represented as

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The superposition states are

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

We now give examples of tensor products in column-vector form:

$$|0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

$$|0\rangle \otimes |+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix},$$

$$|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix},$$

$$|+\rangle \otimes |-\rangle = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}.$$

More generally, suppose Alice has qubit $|A\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle$ and Bob has qubit $|B\rangle = \beta_0 |0\rangle + \beta_1 |1\rangle$. Then their joint state is

$$|A\rangle \otimes |B\rangle = \alpha_0 \beta_0 |0\rangle \otimes |0\rangle + \alpha_0 \beta_1 |0\rangle \otimes |1\rangle + \alpha_1 \beta_0 |1\rangle \otimes |0\rangle + \alpha_1 \beta_1 |1\rangle \otimes |1\rangle.$$

Notice how the multilinearity (bilinearity here) was invoked to obtain this expansion. In column vector form:

$$|A\rangle \otimes |B\rangle = \begin{bmatrix} \alpha_0 \beta_0 \\ \alpha_0 \beta_1 \\ \alpha_1 \beta_0 \\ \alpha_1 \beta_1 \end{bmatrix}.$$

To verify that the composite coefficients are indeed probability amplitudes, we check if the sum of amplitudes squared is 1. But this is easy – according to Quantum Mechanics Law 1, the amplitudes of $|A\rangle$ and $|B\rangle$ satisfy the **normalization condition** i.e. $|\alpha_0|^2 + |\alpha_1|^2 = 1$ and $|\beta_0|^2 + |\beta_1|^2 = 1$, so

$$\begin{aligned} |\alpha_0 \beta_0|^2 + |\alpha_0 \beta_1|^2 + |\alpha_1 \beta_0|^2 + |\alpha_1 \beta_1|^2 &= |\alpha_0|^2 (|\beta_0|^2 + |\beta_1|^2) + |\alpha_1|^2 (|\beta_0|^2 + |\beta_1|^2) \\ &= |\alpha_0|^2 + |\alpha_1|^2 \\ &= 1 \end{aligned}$$

as desired.

Definition 6.2 (Tensor Product of Matrices) Given matrices $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{p \times q}$, their tensor (also called Kronecker) product $A \otimes B$ is the $mp \times nq$ matrix

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix}.$$

Remark 6.3 We seem to have two different (but superficially very similar) notions of tensor products, one for vectors and one for matrices, but note that technically the set of matrices itself is an abstract vector space, thus there really is no difference on an abstract level. The only difference here of course is the *concrete representation* of the objects. What we call vectors in this course we generally write down as columns (long-ish 1D objects), while the operators on these vectors (which technically speaking also form a vector space themselves) we write down as matrices (square-ish 2D objects). Hopefully this isn't too confusing. If it helps you can restack the rectangular-looking matrices into long columns (vectorization). Then the basis matrices spanning matrix space gets rewritten as long, all-zeros-except-one vectors. Then this should be very clear that the tensor product of matrices is exactly the same as the general definition given in [Definition 6.1](#).

Proposition 6.2 The tensor product of matrices satisfies the following properties:

- $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ whenever dimensions match.
- $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$.
- $\text{Tr}(A \otimes B) = \text{Tr}(A)\text{Tr}(B)$.
- If A and B are invertible, then $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

Why are matrix tensor products useful? Recall that matrices appear a lot in quantum, as operations on the quantum states (unitaries, measurements etc). Matrix tensor products characterize the *joint* application of these operations on the entire system. Consider a two qubit basis state $|\psi\rangle = |00\rangle$. Suppose we apply a unitary transformation $U = \begin{bmatrix} p & r \\ q & s \end{bmatrix}$ only to the second qubit.

The overall transformation on $|\psi\rangle$ can be described by

$$|00\rangle \rightarrow |0\rangle \otimes (U|0\rangle)$$

The transformations for the other basis states can be obtained similarly. Specifically:

$$\begin{aligned} |00\rangle &\rightarrow |0\rangle \otimes (U|0\rangle) = \begin{bmatrix} p \\ q \\ 0 \\ 0 \end{bmatrix} \\ |01\rangle &\rightarrow |0\rangle \otimes (U|1\rangle) = \begin{bmatrix} r \\ s \\ 0 \\ 0 \end{bmatrix} \\ |10\rangle &\rightarrow |1\rangle \otimes (U|0\rangle) = \begin{bmatrix} 0 \\ 0 \\ p \\ q \end{bmatrix} \\ |11\rangle &\rightarrow |1\rangle \otimes (U|1\rangle) = \begin{bmatrix} 0 \\ 0 \\ r \\ s \end{bmatrix}. \end{aligned}$$

Denoting I as the identity matrix, this transformation can be succinctly written as

$$I \otimes U |\psi\rangle$$

where

$$I \otimes U = \begin{bmatrix} p & r & 0 & 0 \\ q & s & 0 & 0 \\ 0 & 0 & p & r \\ 0 & 0 & q & s \end{bmatrix}$$

Intuitively, the ‘do nothing’ operation on the first qubit is mathematically represented as applying the identity operator on the first qubit.

Theorem 6.3 (Single Qubit Unitary Transformations in Composite System) Suppose U, V are unitary transformations. If we apply U to $|\phi\rangle$ and V to $|\psi\rangle$, the joint state $|\phi\psi\rangle$ is transformed to $U \otimes V |\phi\psi\rangle$.

Proof. We can consider the unitary transformations applied to $|\phi\rangle$ and $|\psi\rangle$ as two separate unitary transformations applied consecutively:

$$\begin{aligned}(I \otimes V)(U \otimes I) |\phi\psi\rangle &= (IU) \otimes (VI) |\phi\psi\rangle \quad ((A \otimes B)(C \otimes D) = (AC) \otimes (BD)) \\ &= (U \otimes V) |\phi\psi\rangle.\end{aligned}$$

□

Next we consider an extremely important gate in quantum computing, the **CNOT gate**. A CNOT gate takes in two qubits, a control qubit and a target qubit. Denote the control qubit as $|\phi\rangle$ and the target qubit as $|\psi\rangle$.

- If the control qubit is 0, do nothing.
- If the control qubit is 1, flip the target qubit.

Given $|\phi\psi\rangle$, enumerating over all the basis states, we have the following transformations effected by CNOT:

$$\begin{aligned}|00\rangle &\rightarrow |00\rangle \\ |01\rangle &\rightarrow |01\rangle \\ |10\rangle &\rightarrow |11\rangle \\ |11\rangle &\rightarrow |10\rangle.\end{aligned}$$

Note that the resulting basis is still orthonormal. The CNOT gate can be represented by the following matrix,

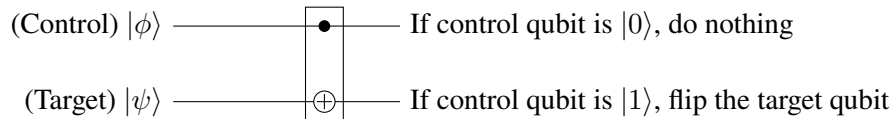
$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

The first two columns in the CNOT matrix represents the transformation to $|\psi\rangle$ when $\phi = 0$, while the last two columns in the CNOT matrix represents the “flipping” transformation to $|\psi\rangle$ when $\phi = 1$.

The probability amplitudes of the original basis will be mapped to the corresponding basis after the CNOT operation. More concretely,

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha_{00} \\ \alpha_{01} \\ \alpha_{10} \\ \alpha_{11} \end{bmatrix} = \begin{bmatrix} \alpha_{00} \\ \alpha_{01} \\ \alpha_{11} \\ \alpha_{10} \end{bmatrix}.$$

CNOT as a diagram:



The next example on creating EPR states is intended to be a hands-on exercise to help you gain familiarity with tensor product manipulation.

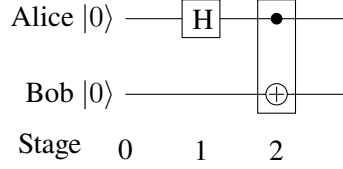
Example 6.4 (Creating EPR/Bell states) To produce the EPR state $|\Phi^+\rangle := \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, we use a circuit consisting of a Hadamard gate (H) followed by a CNOT:

1. **Initial State:** $|\psi_0\rangle = |0\rangle \otimes |0\rangle = |00\rangle$.

2. **Apply H to first qubit:** $|\psi_1\rangle = (H \otimes I) |00\rangle = \left(\frac{|0\rangle+|1\rangle}{\sqrt{2}}\right) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$.

3. **Apply CNOT:** $|\psi_2\rangle = \text{CNOT} \left(\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \right) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

Diagrammatically,



6.2 Quantum Entanglement

Look at the Bell state again. We might think: what is the appropriate $|\psi\rangle$ and $|\phi\rangle$ such that $|\Phi^+\rangle = |\psi\rangle \otimes |\phi\rangle$? That is, what are the individual states of the qubits that collectively are described as the Bell state $|\Phi^+\rangle$ on the entire system?

Answer: there are no pairs of individual states $|\phi\rangle$ and $|\psi\rangle$ such that $|\Phi^+\rangle = |\phi\rangle \otimes |\psi\rangle$.

Proof. Assume that $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is separable, i.e. there exist states $|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle$ and $|\phi\rangle = \beta_0 |0\rangle + \beta_1 |1\rangle$ such that

$$|\Phi^+\rangle = |\psi\rangle \otimes |\phi\rangle.$$

Expanding the tensor product out gives

$$\begin{aligned} |\psi\rangle \otimes |\phi\rangle &= (\alpha_0 |0\rangle + \alpha_1 |1\rangle) \otimes (\beta_0 |0\rangle + \beta_1 |1\rangle) \\ &= \alpha_0 \beta_0 |00\rangle + \alpha_0 \beta_1 |01\rangle + \alpha_1 \beta_0 |10\rangle + \alpha_1 \beta_1 |11\rangle \end{aligned}$$

We compare the coefficients of this expansion to the Bell state $\frac{1}{\sqrt{2}} |00\rangle + 0 |01\rangle + 0 |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$:

1. $\alpha_0 \beta_0 = \frac{1}{\sqrt{2}}$ (from $|00\rangle$)
2. $\alpha_0 \beta_1 = 0$ (from $|01\rangle$)
3. $\alpha_1 \beta_0 = 0$ (from $|10\rangle$)
4. $\alpha_1 \beta_1 = \frac{1}{\sqrt{2}}$ (from $|11\rangle$)

From (2), either $\alpha_0 = 0$ or $\beta_1 = 0$.

- If $\alpha_0 = 0$, then $\alpha_0 \beta_0 = 0$, which contradicts (1).
- If $\beta_1 = 0$, then $\alpha_1 \beta_1 = 0$, which contradicts (4).

Since both possibilities lead to a contradiction, no such α_i, β_i exist. Therefore, $|\Phi^+\rangle$ is nonseparable. $\square$

We say that $|\Phi^+\rangle$ is an **entangled** state. (Quantum) Entanglement is a *core* feature of quantum mechanics and quantum computing, but mathematically it simply expresses the non-separability of the system state.

Definition 6.3 (Separability and Entanglement) Consider a composite quantum system described by the Hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$.

- A pure state $|\Psi\rangle \in \mathcal{H}$ is said to be **separable** (or non-entangled) if it can be written as a tensor product of states from the individual subsystems:

$$|\Psi\rangle = |\psi\rangle_A \otimes |\phi\rangle_B$$

where $|\psi\rangle_A \in \mathcal{H}_A$ and $|\phi\rangle_B \in \mathcal{H}_B$.

- A pure state $|\Psi\rangle$ is said to be **entangled** if it cannot be decomposed in such a manner. That is:

$$|\Psi\rangle \neq |\psi\rangle_A \otimes |\phi\rangle_B \quad \forall |\psi\rangle_A, |\phi\rangle_B$$

While an entangled system state cannot be written as a tensor product of subsystem states, we *can* actually make sense of and talk about these subsystem states, via the concept of partial trace. More on that later in the course.

6.3 Partial Measurements

So far we have seen how to describe composite systems and the operations on them, generally speaking. Performing measurements is also a type of operation. In this section we briefly explore the notion of partial measurements, i.e. making measurements only on a subsystem of the whole system.

Intuitively, if Alice measures *only her qubit* in a joint state, she should obtain a classical outcome (here 0 or 1) according to the Born rule, and the *post-measurement joint state* should be the part of the superposition consistent with that outcome, renormalized to have norm 1.

Consider a general two-qubit state (Alice is the first qubit, Bob is the second):

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle, \quad \sum_{i,j \in \{0,1\}} |\alpha_{ij}|^2 = 1.$$

Alice measures her qubit (in the computational basis). We have the case

$$\Pr[\text{Alice reads } |0\rangle] = |\alpha_{00}|^2 + |\alpha_{01}|^2$$

wherein the state collapses to

$$\frac{\alpha_{00}|00\rangle + \alpha_{01}|01\rangle}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}} = \gamma_0|00\rangle + \gamma_1|01\rangle,$$

where

$$\gamma_0 = \frac{\alpha_{00}}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}}, \quad \gamma_1 = \frac{\alpha_{01}}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}};$$

and the case

$$\Pr[\text{Alice reads } |1\rangle] = |\alpha_{10}|^2 + |\alpha_{11}|^2$$

wherein the state collapses to

$$\frac{\alpha_{10}|10\rangle + \alpha_{11}|11\rangle}{\sqrt{|\alpha_{10}|^2 + |\alpha_{11}|^2}} = \gamma'_0|10\rangle + \gamma'_1|11\rangle,$$

where

$$\gamma'_0 = \frac{\alpha_{10}}{\sqrt{|\alpha_{10}|^2 + |\alpha_{11}|^2}}, \quad \gamma'_1 = \frac{\alpha_{11}}{\sqrt{|\alpha_{10}|^2 + |\alpha_{11}|^2}}.$$

The amplitudes γ_0, γ_1 (respectively γ'_0, γ'_1) are the *conditional* probability amplitudes of the two-qubit basis states given Alice's outcome. Equivalently, they describe Bob's post-measurement state conditioned on Alice:

$$|\psi_B | A = 0\rangle = \gamma_0|0\rangle + \gamma_1|1\rangle, \quad |\psi_B | A = 1\rangle = \gamma'_0|0\rangle + \gamma'_1|1\rangle,$$

since in the $A = 0$ branch the joint state is $|0\rangle \otimes (\gamma_0|0\rangle + \gamma_1|1\rangle)$, and similarly for $A = 1$.

Physically, Alice's measurement *projects* the joint state onto the subspace consistent with her observed classical bit and the renormalization ensures the remaining state has total probability 1. If $|\psi\rangle$ is a product state, measuring Alice does not change Bob's qubit. If $|\psi\rangle$ is entangled, then Bob's conditional state can depend on Alice's outcome, which is the source of the strong correlations observed in entangled measurements.

7 The No-Cloning Theorem and Quantum Teleportation

Every single week during the lecture, a few of you open this overleaf file and copy the scribe template over to your own file to work on it. Some of you use ChatGPT to help refine your scribes (which is okay, just be honest with it). The numerous ctrl-C's/ctrl-V's involved in these undertakings were in all likelihood barely noticed. But they of course were instances of copying/cloning (classical) information, which is ubiquitous. Unfortunately, the same cannot be done with *quantum* information.

7.1 No Quantum Clones

Theorem 7.1 There does not exist a unitary U which upon arbitrary input states $|\psi\rangle$ outputs $|\psi\rangle|\psi\rangle$. More precisely, a unitary such that

$$U|\psi\rangle|\text{anc}\rangle = |\psi\rangle|\psi\rangle \quad \forall |\psi\rangle$$

does not exist. Here $|\text{anc}\rangle$ denotes *any arbitrary* ancillary state which has the same dimension as $|\psi\rangle$.

Proof. We present two proofs.

Proof 1: Suppose there exists such a unitary U . Pick an arbitrary state $|\psi\rangle = a|0\rangle + b|1\rangle$ where a, b are complex amplitudes. Running U on $|\psi\rangle$ yields

$$U(a|0\rangle + b|1\rangle)|\text{anc}\rangle = a^2|00\rangle + ab|01\rangle + ab|10\rangle + b^2|11\rangle.$$

But at the same time we also have $U|0\rangle|\text{anc}\rangle = |00\rangle$ and $U|1\rangle|\text{anc}\rangle = |11\rangle$. By linearity we thus have

$$U(a|0\rangle + b|1\rangle)|\text{anc}\rangle = a|00\rangle + b|11\rangle.$$

Matching the two given expressions for $U|\psi\rangle|\text{anc}\rangle$ we see that they are compatible only when $(a, b) = (0, 1)$ or $(1, 0)$, contradicting the arbitrariness of $|\psi\rangle$.

Proof 2: Pick two arbitrary states $|\psi\rangle, |\phi\rangle$ and run U on them. This gives $U|\psi\rangle|\text{anc}\rangle = |\psi\rangle|\psi\rangle$ and $U|\phi\rangle|\text{anc}\rangle = |\phi\rangle|\phi\rangle$. Taking their inner products and noting the unitarity of U we see that $\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2 \implies \langle\psi|\phi\rangle = 0/1$. That is, the two states must either be the same or mutually orthogonal, once again contradicting the assumed arbitrariness. $\square$

7.2 Quantum Teleportation

Consider the following scenario: suppose we have 2 friends, Alice and Bob, and they live far away from one another. Alice has in her possession some unknown qubit $|\psi\rangle := \alpha|0\rangle + \beta|1\rangle$ and she wants to send it to Bob. How can she do it?

7.2.1 A Simpler Puzzle

Before we start off showing how it can be done, let us start off with a simple puzzle. Suppose Alice and Bob are sitting in the lab, and Alice wants to send some unknown qubit $|\psi\rangle$ to Bob. Obviously, she cannot walk up to Bob to give it to him but what they do have is a shared network for both of them to perform unitary operations (see [Figure 1](#)).

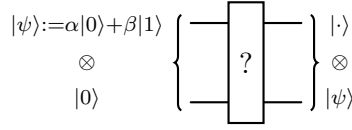


Figure 1: Some unknown quantum teleporter

Note that from Figure 1, we want these states to be un-entangled so that Alice cannot “destroy” the qubit via a measurement.

What Alice can do is to apply a CNOT operation with a control qubit $|0\rangle$ followed by a Hadamard operation as follows in Figure 2:

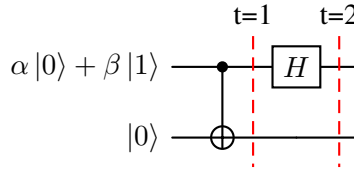


Figure 2: Alice’s CNOT and Hadamard Operation

The intuition here is this: suppose Alice measures her qubit, the quantum state of the qubit is going to collapse to $|0\rangle$ or $|1\rangle$, which is clearly not what we want. By introducing the CNOT gate, the state is now entangled. One can realise that neither qubit in the entangled state contains the full information of the original qubit. Therefore at $t = 1$, the state of the quantum system, $|\psi_1\rangle$ is:

$$|\psi_1\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle = \alpha|00\rangle + \beta|11\rangle \quad (7.1)$$

Now, we want to remove the entanglement of the quantum state. Alice *could have* measured her qubit but that is not going to be helpful because with probability $|\alpha|^2$, her qubit is going to be $|0\rangle$ but then due to the entanglement, Bob’s qubit is going to be $|0\rangle$ with the same probability. The same explanation applies for the state $|1\rangle$ with probability $|\beta|^2$. Applying the Hadamard operation on only Alice’s qubit can remove that entanglement. We can see that at $t = 2$, the state of the quantum system $|\psi_2\rangle$ is

$$\begin{aligned} |\psi_1\rangle &\rightarrow |\psi_2\rangle \\ &= \alpha|+0\rangle + \beta|-1\rangle \\ &= \frac{\alpha}{\sqrt{2}}|00\rangle + \frac{\alpha}{\sqrt{2}}|10\rangle + \frac{\beta}{\sqrt{2}}|01\rangle - \frac{\beta}{\sqrt{2}}|11\rangle \end{aligned} \quad (7.2)$$

Now, when Alice does a partial measurement on her own qubit, the quantum state will collapse to

$$\begin{cases} \alpha|00\rangle + \beta|01\rangle & w.p. \frac{|\alpha|^2 + |\beta|^2}{2} = \frac{1}{2} \\ \alpha|10\rangle - \beta|11\rangle & w.p. \frac{1}{2} \end{cases} \quad (7.3)$$

where the first case occurs when Alice observes $|0\rangle$ whereas the second case occurs when Alice observes $|1\rangle$. Denote the bit (not qubit anymore for Alice) that Alice observes to be b .

Alice then sends b to Bob (perhaps by telling Bob due to their close proximity in the lab). If $b = 0$, as mentioned above, the quantum state must have collapsed to $\alpha|00\rangle + \beta|01\rangle$. Bob’s qubit would have been $\alpha|0\rangle + \beta|1\rangle$, which is exactly $|\psi\rangle$! However, if $b = 1$, then the quantum state must have collapsed to $\alpha|10\rangle - \beta|11\rangle$. Bob’s qubit is $\alpha|0\rangle - \beta|1\rangle$. Bob simply has to perform a unitary operation, $U = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ to obtain $|\psi\rangle$!

7.2.2 Quantum Teleportation with an EPR Pair

Now suppose that Alice and Bob are not near one another anymore but instead are really far apart! However, before Bob left Alice, both of them share an EPR pair. That is, each of them *share a qubit* of the EPR pair.

The first step is for Alice to entangle $|\psi\rangle$ with her own EPR pair using the CNOT and Hadamard operation as before. Bob's qubit (the EPR pair) remains untouched. The step involved is shown in [Figure 3](#)

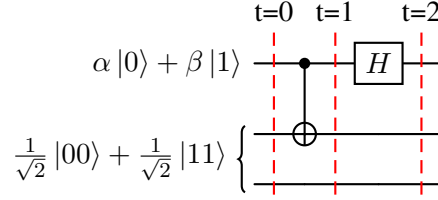


Figure 3: Entanglement of Alice's EPR pair with her qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

Let us go through the states for each of the time steps. At the beginning, $t = 0$, the joint state of the quantum circuit is given by $|\psi_0\rangle$:

$$|\psi_0\rangle = \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle \quad (7.4)$$

At $t = 1$, Alice has entangled her qubit with her own EPR pair to get a new joint state, $|\psi_1\rangle$:

$$|\psi_1\rangle = \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|110\rangle + \frac{\beta}{\sqrt{2}}|101\rangle \quad (7.5)$$

At $t = 2$, Alice has passed her qubit through a Hadamard gate which now results in $|\psi_2\rangle$

$$\begin{aligned} |\psi_1\rangle &= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|110\rangle + \frac{\beta}{\sqrt{2}}|101\rangle \\ &\downarrow \\ |\psi_2\rangle &= \frac{\alpha}{\sqrt{2}}|+00\rangle + \frac{\alpha}{\sqrt{2}}|+11\rangle + \frac{\beta}{\sqrt{2}}|-10\rangle + \frac{\beta}{\sqrt{2}}|-01\rangle \\ &= \frac{\alpha}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}(|000\rangle + |100\rangle) \right) + \frac{\alpha}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}(|011\rangle + |111\rangle) \right) \\ &\quad + \frac{\beta}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}(|010\rangle - |110\rangle) \right) + \frac{\beta}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}(|001\rangle - |101\rangle) \right) \\ &= \frac{\alpha}{2}(|000\rangle + |100\rangle + |011\rangle + |111\rangle) + \frac{\beta}{2}(|010\rangle + |001\rangle - |110\rangle - |101\rangle) \end{aligned} \quad (7.6)$$

The next crucial step for Alice is to do a partial measurement on the 2 qubits and *communicate her observations*, b_0b_1 , to Bob. The measurement should result in an observation of one of 4 possible states: $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$. The effect of the measurements are detailed in [Table 1](#).

Now let's turn to Bob. Suppose he receives Alice's 2-bit observation. He can then perform the following unitary operations according to the observed b_0b_1 to *directly recover* $|\psi\rangle$. The unitary operation U , which Bob has to

Alice's observation, $b_1 b_0$	State Collapses to	Bob's State
00	$\alpha 000\rangle + \beta 001\rangle$	$ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$
01	$\alpha 011\rangle + \beta 010\rangle$	$\alpha 1\rangle + \beta 0\rangle$
10	$\alpha 100\rangle - \beta 101\rangle$	$\alpha 0\rangle - \beta 1\rangle$
11	$\alpha 111\rangle - \beta 110\rangle$	$\alpha 1\rangle - \beta 0\rangle$

Table 1: Alice's observation and the effects of her measurement on the quantum circuit and hence, Bob's state.

perform, is as follows (note the order of operations):

$$U = \begin{cases} I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ (equivalently: do nothing)} & \text{if } b_0 b_1 = 00 \\ X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \text{if } b_0 b_1 = 01 \\ Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} & \text{if } b_0 b_1 = 10 \\ ZX = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} & \text{if } b_0 b_1 = 11 \end{cases}$$

We finally give the full diagram of the quantum circuit in Figure 4.

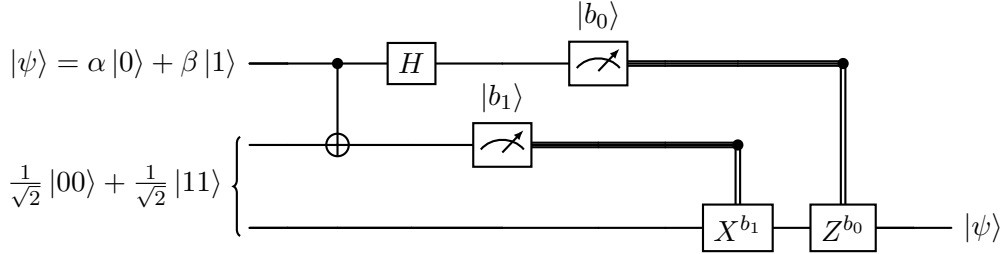


Figure 4: Quantum Teleportation Circuit with EPR Pair.

Remark 7.1 It might seem as if the teleported $|\psi\rangle$ is being cloned, which violates the Non-Cloning theorem. This is in fact not the case because Alice has lost her own qubit! The No-Cloning theorem is henceforth not violated. A useful sanity check is to realise that all the operations that we have done so far are linear.

In conclusion, quantum teleportation demonstrates how quantum information can be transferred between distant locations without physically sending the particle that carries the state. By combining entanglement, measurement, and a bit of classical communication overhead, the protocol faithfully and securely reconstructs an unknown quantum state at a remote site (while necessarily destroying the original). This process highlights the power of entanglement as a resource and forms a foundational building block for quantum communication, quantum networks, and future quantum internet technologies.

Summary

Idea: Alice has a state $|\psi\rangle$ (whose amplitudes she doesn't even know). Through the quantum teleportation protocol she can send it to Bob using a shared $|EPR\rangle$ and two bits of classical communication.

Protocol: Starting from $|\psi\rangle |EPR\rangle$, Alice applies $H \circ CNOT$ on her two qubits to obtain

$$\frac{1}{2} [|00\rangle (\alpha |0\rangle + \beta |1\rangle) + |01\rangle (\alpha |1\rangle + \beta |0\rangle)]$$

$$+ |10\rangle (\alpha |0\rangle - \beta |1\rangle) + |11\rangle (\alpha |1\rangle - \beta |0\rangle)].$$

Alice then measures her two qubits and obtains the results (b_0, b_1) . Alice sends her results (comprising two bits of information) to Bob, who acts accordingly (apply $Z^{b_0} X^{b_1}$ on his qubit) to recover $|\psi\rangle$.

8 Classical Computation $\subseteq$ Quantum Computation

KK: Add the material on reversible computing, toffoli, quantum circuits computing boolea functions etc. It seems that these are not yet fully covered from last lecture.

KK: TBD

Instructions

- Try to leave the preamble unchanged, only make changes to `main.tex` and `references.bib`. Add images to the `images` folder.
- Share one (zipped) directory preserving the directory structure on Canvas.
- Prepare organized notes by elaborating on the broad items provided in different sections below based on the lecture material completing any missing details from lectures.
- Feel free to modify structure and/or order of contents as you see fit; the below is just a guideline.

9 Basic Primitives in Quantum Computing

Add notion and details of ‘uncomputing’. We uncompute in order to be able to reuse ancillas.

Add notion and details of the ‘phase-kickback’ method.

Add details on the Hadamard-Walsh transform, essentially is parallel Hadamards.

Instructions

- Try to leave the preamble unchanged, only make changes to `main.tex` and `references.bib`. Add images to the `images` folder.
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- Feel free to modify structure and/or order of contents as you see fit; the below is just a guideline.

10 Foundational Quantum Algorithms

KK: For all material here, add/remove details as you see fit. Please add quantum circuits if available, use the `quantikz` package.

10.1 Quantum Fourier Transform

1. Binary representation: if $\alpha \geq 1$, $\alpha = \alpha_1 \dots \alpha_n = \alpha_1 2^{n-1} + \dots + \alpha_n 2^0$; if $0 \leq \alpha < 1$, $\alpha = 0.\alpha_1 \dots \alpha_n = \alpha_1 2^{-1} + \dots + \alpha_n 2^{-n}$. Here, $\alpha_i \in \{0, 1\}$ for all i . Note that if $\alpha = \alpha_1 \dots \alpha_n$, then $\alpha/2^n = 0.\alpha_1 \dots \alpha_n$.
2. Let $\mathbf{x} \in \mathbb{C}^N$ (recall for simplicity we take $N = 2^n$ for some $n \in \mathbb{N}$). The DFT (discrete Fourier transform) implements $\mathbf{x} \mapsto \mathbf{y} = \text{DFT}(\mathbf{x})$, where

$$\text{DFT} = \frac{1}{\sqrt{N}} \sum_{j,k=0}^{N-1} e^{2\pi i j k / N} |j\rangle\langle k|.$$

The QFT implements the same transformation: for a basis state $|j\rangle$ we have

$$\text{QFT} |j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i j k / N} |k\rangle.$$

In binary, this is

$$\text{QFT} |j_1 \dots j_n\rangle = \frac{1}{\sqrt{2^n}} (|0\rangle + e^{2\pi i 0 \cdot j_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot j_{n-1} j_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0 \cdot j_1 j_2 \dots j_n} |1\rangle).$$

3. Fast Fourier Transform: Cooley-Tukey Algorithm. Start from

$$\begin{aligned} y_k &= \sum_{j=0}^{N-1} e^{\frac{2\pi i j k}{N}} x_j \\ &= \sum_{j=0}^{N/2-1} e^{\frac{2\pi i (2j)k}{N}} x_{2j} + \sum_{j=0}^{N/2-1} e^{\frac{2\pi i (2j+1)k}{N}} x_{2j+1} \end{aligned}$$

$$= \underbrace{\sum_{j=0}^{N/2-1} e^{\frac{2\pi i j k}{N/2}} x_{2j}}_{E_k} + e^{\frac{2\pi i k}{N}} \underbrace{\sum_{j=0}^{N/2-1} e^{\frac{2\pi i j k}{N/2}} x_{2j+1}}_{O_k}.$$

For brevity we have absorbed $1/\sqrt{N}$ into the x_i . The key is to notice that

$$y_{k+\frac{N}{2}} = E_k - e^{\frac{2\pi i k}{N}} O_k$$

so we only have to compute y_k from $k = 0$ until $k = N/2 - 1$. We recursively break this up into smaller DFTs until we reach the base case of $N = 2$. Essentially, the key breakthrough of Cooley-Tukey is to reuse results of intermediate computations to compute the y_k 's. One can see total complexity is $O(N \log N)$ by drawing the binary tree expansion of this procedure: at each level l , there are $2^l E_k/O_k$ terms, and for each such term we have $N/2^l k$'s. Each E_k/O_k term is the sum of its children nodes, so takes $O(1)$ computations to evaluate. So each level takes $O(N)$ operations, and there are $\log N$ levels.

4. Derivation of binary representation of QFT:

$$\begin{aligned} \text{QFT } |j_1 \dots j_n\rangle &= \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{\frac{2\pi i j k}{2^n}} |k\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{k_1, \dots, k_n=0}^1 e^{2\pi i j (\sum_{l=1}^n k_l 2^{-l})} |k_1 \dots k_n\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{k_1, \dots, k_n=0}^1 \bigotimes_{l=1}^n e^{2\pi i j k_l 2^{-l}} |k_l\rangle \\ &= \frac{1}{\sqrt{2^n}} \bigotimes_{l=1}^n \left[\sum_{k_l=0}^1 e^{2\pi i j k_l 2^{-l}} |k_l\rangle \right] \\ &= \frac{1}{\sqrt{2^n}} \bigotimes_{l=1}^n \left[|0\rangle + e^{2\pi i j 2^{-l}} |1\rangle \right] \\ &= \frac{1}{\sqrt{2^n}} (|0\rangle + e^{2\pi i 0 \cdot j_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot j_{n-1} j_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0 \cdot j_1 j_2 \dots j_n} |1\rangle). \end{aligned}$$

This representation gives intuition toward constructing an efficient circuit for the QFT. With the phase gates $R_k := \text{diag}(1, e^{\frac{2\pi i}{2^k}})$, and M_i meaning single-qubit gate M acts on qubit i , the circuit comprises first Hadamarding then applying ctrl- R_k 's, see [NC10] pg219 or wiki.

5. Speedup: FFT takes $\Theta(N \log N)/\Theta(n2^n)$, QFT takes $\Theta(\log^2 N)/\Theta(n^2)$. Caveat: the Fourier-transformed output QFT $|x\rangle$ is not DFT($\mathbf{x}$) itself, rather QFT $|x\rangle$ has the entries of DFT($\mathbf{x}$) encoded in its amplitudes. To extract the values of the amplitudes we generally have to perform measurements which generally consumes more resources.

10.2 The Deutsch-Jozsa Family

First consider the baby version, 'Deutsch'. 'Deutsch-Jozsa' is a generalization of Deutsch to n -input bit functions.

Deutsch

1. Idea: Determine whether a one-bit function f is constant or balanced. f is implemented through the oracle $O_f |x, y\rangle = |x, y \oplus f(x)\rangle$ which is a permutation and hence also a unitary.

2. Method: Start from $|01\rangle$, apply $O_f \circ H \otimes H$ to get (up to a global phase) $|\pm\rangle|-\rangle$ (+ if $f(0) = f(1)$, - if $f(0) \neq f(1)$). Discard second qubit, Hadamard the first, so we get $|0\rangle \iff f(0) = f(1) \iff f(0) \oplus f(1) = 0$ (constant), $|1\rangle \iff f(0) \neq f(1) \iff f(0) \oplus f(1) = 1$ (balanced). Only requires one O_f call.

3. Note that key is $O_f |x\rangle|-\rangle = (-1)^{f(x)} |x\rangle|-\rangle$, ‘phase-kickback’.

Deutsch-Jozsa: Generalization of Deutsch to n -bit functions. Leverages massively parallel phase-kickback.

1. Note that $H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_y (-1)^{x \cdot y} |y\rangle$.
2. Start from $|0^n\rangle|1\rangle$, apply $H^{\otimes n} \circ O_f \circ H^{\otimes n} \otimes H$ to get the state

$$\sum_{z \in \{0,1\}^n} \left(\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x) + x \cdot z} \right) |z\rangle.$$

Then measure the working n qubits. Observe that $P(z = 0^n) = \left| \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \right|^2$ equals 1 iff f is constant and equals 0 iff f is balanced. Again, only requires one O_f call.

3. Classically, deterministic worst case needs $2^{n-1} + 1 = O(2^n)$ queries. Randomized algorithm: randomly pick k distinct inputs x_i and query to get $f(x_i)$, answer constant if $f(x_1) = \dots = f(x_k)$, balanced otherwise. If f was really constant then method works with certainty, if f was balanced then $f(x_1) = \dots = f(x_k)$ occurs with probability $\frac{1}{2^{k-1}}$. So if probability-error tolerance is ε , need $k = O(\log \frac{1}{\varepsilon})$.

Related to DJ is Bernstein-Vazirani, albeit with a slightly different end goal. Algorithm similar.

Bernstein-Vazirani: Given $f : \{0,1\}^n \rightarrow \{0,1\}$ s.t. $f(x) = x \cdot s$ for some unknown s , find s .

1. Classically, most efficient method is to obtain $f(1,0,\dots) = s_1, f(0,1,\dots) = s_2, \dots$. This requires n queries.
2. Quantumly, the algorithm is the same as DJ: apply $H^{\otimes n} \circ O_f \circ H^{\otimes n} \otimes H$. Miraculously, the final output is $|s\rangle$!
3. Ok actually nothing miraculous - it's simply because after applying the query, we get the state $\frac{1}{\sqrt{2^n}} \sum_x (-1)^{x \cdot s} |x\rangle$, which is $H^{\otimes n} |s\rangle$. And Hadamards are self-inverse.

10.3 Simon's Problem and Algorithm

1. Problem: given $f : \{0,1\}^n \rightarrow \{0,1\}^n$ s.t. $f(x) = f(y) \iff x = y \oplus s$, find s .
2. Classical Case

- (a) A randomized algorithm: randomly pick k (TBD) distinct inputs x_i and query to get $f(x_i)$. If there is no collision, answer $s = 0^n$. If there is a collision $f(x_i) = f(x_j)$, then answer $s = x_i \oplus x_j$.
- (b) Analysis: If actually $s = 0^n$ then the algorithm works with certainty. If actually $s \neq 0^n$ then we might answer wrongly if there is no observed collisions. Let the probability that the samples have no collision after $k \geq 2$ queries be $P(k)$. Suppose after $j-1$ queries there is no matching. The probability that on the j th, i.e. next query there is still no collision with one of the previous queries is then $1 - \frac{j-1}{2^n - (j-1)}$. Thus after k rounds,

$$P(k) = \prod_{j=2}^k \left[1 - \frac{j-1}{2^n - (j-1)} \right]$$

$$\geq 1 - \sum_{j=2}^k \left[\frac{j-1}{2^n - (k-1)} \right]$$

$$\geq 1 - \frac{\Theta(k^2)}{2^n - \Theta(k)}.$$

Remember that we *do* want a collision! So we want $P(k)$ to be small, say ε (equivalently, coll. prob. is $1 - \varepsilon$). Then $P(k) < \varepsilon \implies k \in \Omega(2^{n/2})$. Here in the first $\geq$ we used $(1-a)(1-b) \geq 1 - (a+b)$ for $0 \leq a, b \leq 1$. Also note that we can assume from the outset that $k \leq 2^{n/2}$ because if $k > 2^{n/2}$ then there has to be a collision.

- (c) Simon also showed *any* randomized algorithm needs $k = \Omega(2^{n/2})$ queries (proof is similar to the one above).

3. Simon's Quantum Algorithm

- (a) Quantum part: Start with n work qubits and n ancilla qubits $|0^n\rangle |0^n\rangle$. Apply $(H^{\otimes n} \otimes I)O_f(H^{\otimes n} \otimes I)$ then measure the work qubits.
- (b) Classical postprocessing: repeat $n-1$ times to get $n-1$ strings y_i all satisfying $y_i \cdot s = 0$, where y_i are linearly independent with positive probability. If lin. ind., use Gaussian elimination to solve system, giving two candidates 0^n and $s' \neq 0^n$.
- (c) Test: if $f(0^n) = f(s')$, answer $s = s'$; otherwise $f(0^n) \neq f(s')$, answer $s = 0^n$.
- (d) Boosting: repeat entire process k times to boost prob. of obtaining lin. ind. y_i . (LI can be easily checked.) If for any run $f(0^n) = f(s')$, answer $s = s'$; otherwise if for all runs $f(0^n) \neq f(s')$, answer $s = 0^n$.

4. Analysis of Simon's algorithm

- (a) Quantum: after applying the unitaries the state is

$$|\psi\rangle = \sum_y |y\rangle \left(\frac{1}{2^n} \sum_x (-1)^{x \cdot y} |f(x)\rangle \right).$$

If $s = 0^n$, f is injective so $p(y) = \frac{1}{2^n}$ for all y . If $s \neq 0^n$, f is two-to-one, so if $z \in \text{range}(f)$ then $f(x_z) = f(x'_z) = z$ for $x_z = x'_z \oplus s$. Then $p(y) = \frac{1}{2^{n-1}}$ if $y \cdot s = 0$, and $p(y) = 0$ if $y \cdot s = 1$.

- (b) Postprocessing: suppose we have lin. ind. $\{y_i\}_{i=1}^{n-1}$. GE gives the two possible solutions: $0^n, s' \neq 0^n$. Suppose actually $s = 0^n$. Then $f(0^n) \neq f(s')$, so our answer $s = 0^n$ is always correct. Suppose actually $s \neq 0^n$. Then $f(s) = f(0^n) = f(s')$, so our answer $s' = s$ is always correct.
- (c) Probability of linear independence: given m LI strings $y_1, \dots, y_m$, spanning subspace is size 2^m . Prob of y_{m+1} being LI to this is $\frac{2^n - 2^m}{2^n}$. So prob of $n-1$ LI strings is $P = (1 - \frac{2^1}{2^n}) \dots (1 - \frac{2^{n-2}}{2^n}) \geq 1 - \sum_{i=1}^{n-2} \frac{1}{2^i} > \frac{1}{2}$. So prob. of *no* LI even after running k times is at most $(1 - \frac{1}{2})^k < e^{-k/2}$. So prob. of *at least one* LI after running k times is at least $1 - e^{-k/2}$.

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